THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 2

Chapter 10: **The Human Eye and the Colourful World** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Spectacles for hypermetropia use 1

(a) convex lenses **(b)** concave lenses **(c)** prisms **(d)** plane lenses

2. Which colour has the longest wavelength in the visible spectrum? 1

(a) Red **(b)** Green **(c)** Violet **(d)** Blue

3. The eye lens is made of 1

(a) plastic **(b)** opaque tissue

(c) a transparent, flexible material **(d)** glass

4. Which of these is not a defect of vision? 1

(a) Presbyopia **(b)** Hypermetropia **(c)** Accommodation **(d)** Myopia

5. The Sun's white light is made of 1

(a) seven colours **(b)** one colour

(c) only red and blue **(d)** no colours

6. The nerve that carries signals to the brain is the 1

(a) sciatic nerve **(b)** ciliary nerve **(c)** optic nerve **(d)** spinal nerve

7. Cataract can be cured by 1

(a) convex lens **(b)** concave lens **(c)** surgery **(d)** prism

8. The Tyndall effect is seen in 1

(a) colloidal solutions **(b)** pure water

(c) solid metals **(d)** vacuum

9. The cells in the retina that respond to light are 1

(a) red blood cells **(b)** neurons only **(c)** muscle cells **(d)** rods and cones

10. The power of a concave lens is 1

(a) negative **(b)** infinite **(c)** zero **(d)** positive

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Name the defect of vision corrected by a convex lens. 2

12. What is the Tyndall effect? 2

13. What is the near point of a normal eye? 2

14. What is presbyopia? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. What is accommodation of the eye? How does the lens change for near objects? 3

16. Write the order of the seven colours of white light in the spectrum. 3

17. Why is red used for danger signals? 3

OR

A lens has a focal length of -0.5 m. Find its power and its nature. 3

18. What is atmospheric refraction? Name two phenomena caused by it. 3

OR

What is scattering of light? Give an example. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Define myopia. State its causes and correction. 5

OR

Explain why the sky is blue and the Sun appears red at sunrise and sunset. 5

20. Define hypermetropia. State its causes and correction. 5

OR

Describe the Tyndall effect with two examples and explain. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND